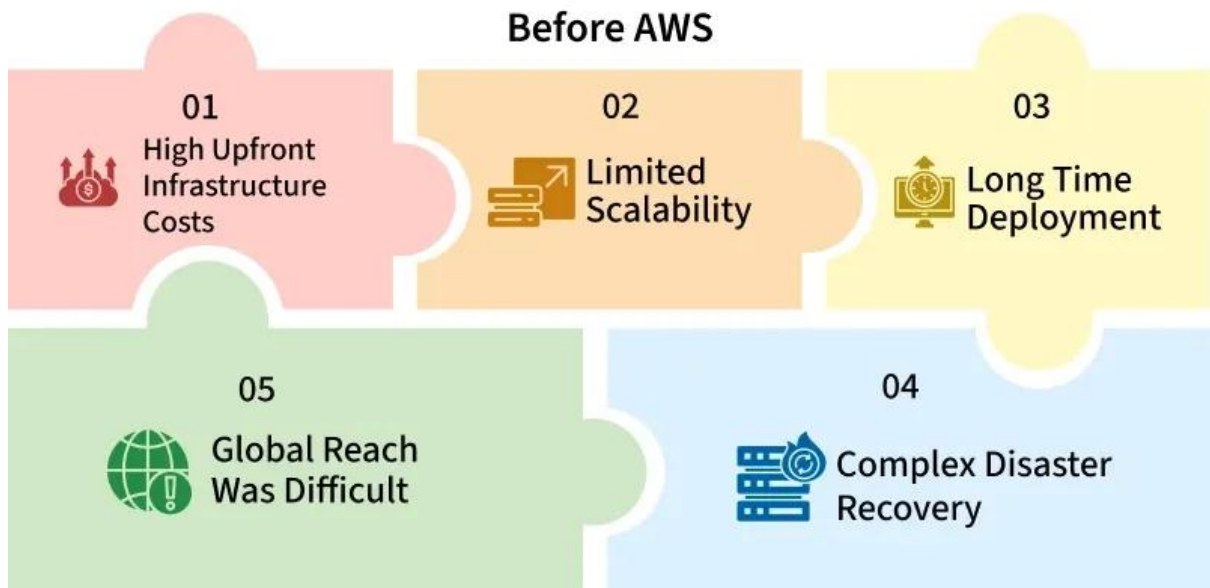
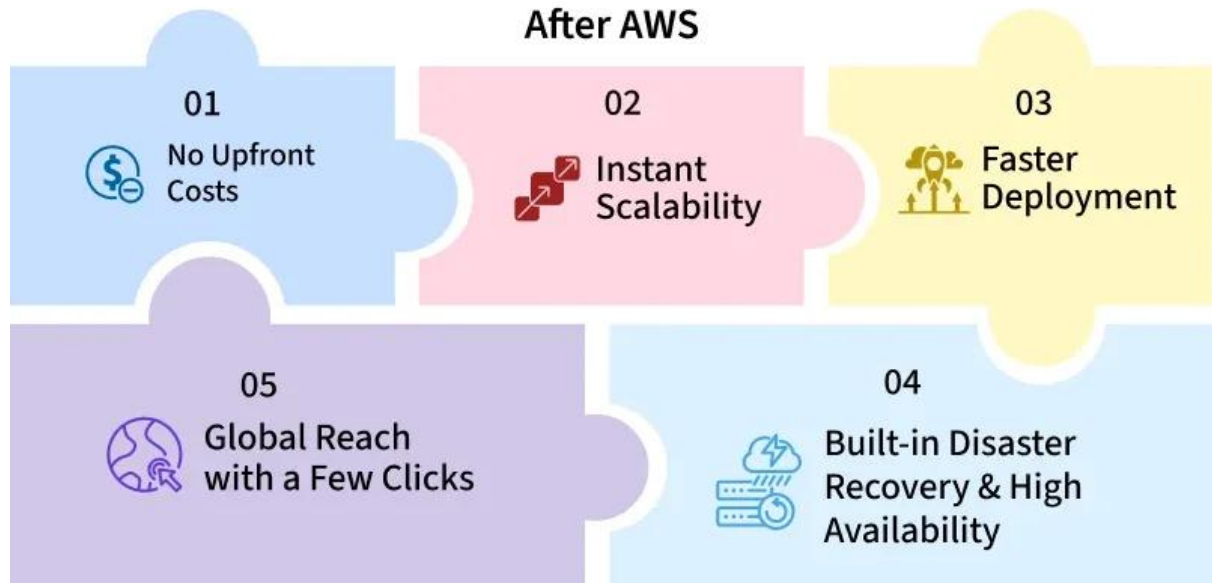


AWS Introduction

Access technology services such as computing power, storage, and databases on an on-demand basis.



After AWS



The AWS Global Infrastructure

The backbone of AWS is its massive global infrastructure. Understanding this is key to architecting resilient applications.

1. Regions: A Region is a physical location in the world where AWS clusters data centers. (e.g., us-east-1 in N. Virginia or ap-south-1 in Mumbai).

Why it matters: You choose a region to optimize latency (closeness to users), minimize costs (prices vary by region), and meet data sovereignty/compliance laws.

2. Availability Zones (AZs): Each Region consists of multiple, isolated locations known as Availability Zones.

An AZ is one or more discrete data centers with redundant power, networking, and connectivity.

Why it matters: If you run your app in just one data center and it floods, your app dies. If you run it across multiple AZs, your app stays online even if one building fails.

3. Edge Locations: These are smaller data centers located in major cities worldwide, used primarily for Amazon CloudFront (CDN) to cache content closer to end-users to reduce latency.

AWS Services

AWS offers over 200 services. Here are the fundamental ones you must know to get started.

1. Compute Services

Amazon EC2 (Elastic Compute Cloud): Resizable virtual servers. You choose the OS (Linux/Windows) and the hardware power (CPU/RAM). It is the workhorse of AWS.

AWS Lambda: Serverless compute. You upload your code, and Lambda runs it only when triggered (by an event like a file upload). You don't manage any servers.

AWS Elastic Beanstalk: Platform as a Service (PaaS) for deploying web apps. You upload code, and AWS handles the deployment (capacity provisioning, load balancing, auto-scaling).

2. Storage Services

Amazon S3 (Simple Storage Service): Object storage for files (images, videos, backups). It allows you to store and retrieve any amount of data from anywhere on the web.

Amazon EBS (Elastic Block Store): Block storage (virtual hard drives) that you attach to EC2 instances. It is persistent storage for your virtual servers.

Amazon Glacier: Extremely low-cost storage for data archiving and long-term backup.

3. Database Services

Amazon RDS (Relational Database Service): Managed SQL databases. It supports engines like MySQL, PostgreSQL, Oracle, SQL Server, and Amazon Aurora (AWS's high-performance proprietary engine).

Amazon DynamoDB: A managed NoSQL database service that provides fast and predictable performance with seamless **scalability**.

4. Networking Services

Amazon VPC (Virtual Private Cloud): Lets you provision a logically isolated section of the AWS Cloud where you can launch resources in a virtual network you define.

Amazon CloudFront: A Content Delivery Network (CDN) that speeds up distribution of your static and dynamic web content to users.

Amazon Route 53: A highly available and scalable cloud Domain Name System (DNS) web service.